Study Breaks, Perceived Study Efficiency, and Next-Day Retention

STA304H5 | Research Proposal | Fall 2026

Research Objective

This study examines study-break habits and perceived learning outcomes among Fall 2026 STA304H5 students. We focus on break frequency, average duration, and exercise/gaming activity profiles. Perceived efficiency is measured by agreement that breaks improve efficiency in the next study period. Perceived retention is measured by a self-reported next-day recall rating.

Research Questions

1. Is break frequency associated with students' agreement that breaks improve their efficiency in the next study period?

2. Is average break duration linearly associated with perceived next-day retention among students who take study breaks?

3. Does mean perceived next-day retention differ across four mutually exclusive exercise/gaming activity profiles among students who take study breaks?

Population, Sampling, and Data Collection

The target population is students enrolled in STA304H5 in Fall 2026. A Google Forms questionnaire was distributed through the course survey exchange platform, with reciprocal survey completion encouraging participation. Recruitment produced a voluntary-response sample. All 57 submissions are retained in the source dataset; eligibility and data-quality screening are applied separately for analysis.

The questionnaire collects study-session length, number of breaks, average break duration, weekly study hours, break activities, and student background information. Retention is recorded on a 10-point scale and agreement about efficiency on a seven-point scale. Break activities allow multiple selections.

Statistical Analysis

For RQ1, we will use simple linear regression with the seven-point efficiency agreement score as the response and break count as the predictor. For RQ2, we will regress the retention rating on average break duration in minutes among break takers. Each slope will be tested against zero, with a 95% confidence interval reported.

For RQ3, we will use one-way ANOVA to compare mean retention across four groups: exercise without gaming, gaming without exercise, both, and neither. Other activities may co-occur in any group. Restricting the comparison to these two activity selections assigns each break taker to exactly one group and avoids overlapping ANOVA groups. The null hypothesis is equality of the four population means. Tests will use a 5% significance level; raw and Holm-adjusted p-values will be reported for the three research questions.

The primary analysis will exclude two records with unavailable course IDs and seven with reported total break time at least as long as the study session, leaving 48 respondents. Other IDs remain unverified. RQ2 uses 45 complete break takers; RQ3 uses 46 break takers. Four impossible weekly-hour values are treated as missing in that variable only. Unresolved ranges remain missing, and break duration is undefined for respondents reporting no breaks.

We will summarize variables, plot the relationships, and examine residual normality, variance and influential observations; regression diagnostics will also assess linearity. Sensitivity analyses will restore timing-flagged records and assess the influential long-break response. Screening decisions and analysis-specific sample sizes will be documented.

Scope and Limitations

Ratings will be treated as approximately numerical. Both outcomes reflect perceptions rather than measured performance. Voluntary participation, record exclusions and small activity groups may limit precision and generalizability. Activity comparisons do not adjust for session length or other potential confounders. The observational design supports associations, not causal effects.

Questionnaire: Google Form